WORKED SOLUTION

STEP I 2004, Question 8

Pure • Monotone Sequences and Induction

STEP I 2004, Question 8

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/04-s1-q8

The key idea

To show a sequence is increasing, look at the difference $t_{n+1} - t_n$ as a function of t_n. In both parts it factorises, and the factors are exactly the fixed points 2 and 3 of the recurrence (the values where $t_{n+1} = t_n$). The sign of the difference then depends only on where t_n lies relative to 2 and 3, so the real work is an induction showing that every term stays in the right region.

Solution

Part (i)

For any n,

$$x_{n+1} - x_n = \frac{x_n^2 + 6}{5} - x_n = \frac{x_n^2 - 5x_n + 6}{5} = \frac{(x_n - 2)(x_n - 3)}{5}.$$

If $x_n > 3$ then both factors are positive, so $x_{n+1} - x_n > 0$, that is, $x_{n+1} > x_n$.

It remains to check that *every* term exceeds 3, which we do by induction. The statement $x_n > 3$ holds for $n = 0$ by hypothesis. Suppose $x_n > 3$. By the calculation above, $x_{n+1} > x_n > 3$, so $x_{n+1} > 3$ too. Hence $x_n > 3$ for all $n \geq 0$, and therefore $x_{n+1} > x_n$ for all $n \geq 0$: the sequence is strictly increasing.

? **Ask yourself** Why isn't it enough to show $x_1 > x_0$?

The inequality $x_{n+1} > x_n$ was deduced from $x_n > 3$, so we need that condition at every step, not just the first. Here it comes for free: an increasing step from above 3 stays above 3. (Note that starting at exactly $x_0 = 3$ gives the constant sequence, so the strict inequality matters.)

Part (ii)

First note that if $2 < y_n < 3$ then $y_n > 0$, so $y_{n+1} = 5 - 6/y_n$ is defined. For such y_n,

$$y_{n+1} - y_n = 5 - \frac{6}{y_n} - y_n = -\frac{y_n^2 - 5y_n + 6}{y_n} = \frac{(y_n - 2)(3 - y_n)}{y_n},$$

which is positive, since $y_n - 2 > 0$, $3 - y_n > 0$ and $y_n > 0$. So $y_{n+1} > y_n$ *provided* $2 < y_n < 3$.

As in part (i), we must show that every term lies in this interval. This time it is not automatic, because an increasing sequence could overshoot 3. So we look at the distance from 3:

$$3 - y_{n+1} = 3 - 5 + \frac{6}{y_n} = \frac{6 - 2y_n}{y_n} = \frac{2(3 - y_n)}{y_n}.$$

? **Ask yourself** How do we know the sequence never jumps past 3?

The formula above says that $3 - y_{n+1}$ has the same sign as $3 - y_n$ whenever $y_n > 0$. So a term below 3 is always followed by a term below 3.

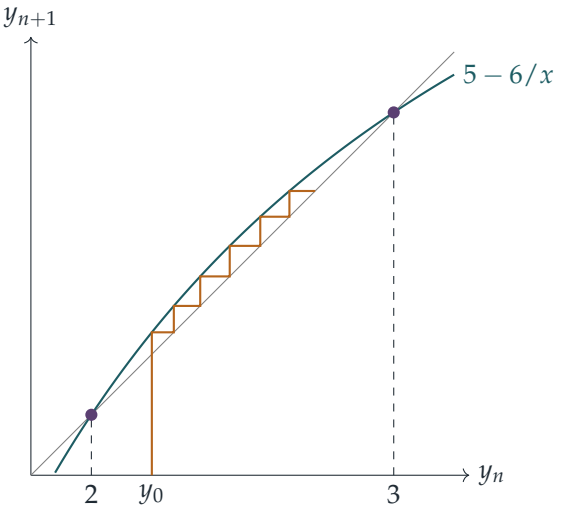
Now the induction. Claim: $2 < y_n < 3$ for all $n \geq 0$. This holds for $n = 0$ by hypothesis. Suppose $2 < y_n < 3$. Then:

- $y_{n+1} > y_n > 2$, by the difference calculation;
- $3 - y_{n+1} = 2(3 - y_n)/y_n > 0$, since $3 - y_n > 0$ and $y_n > 0$, so $y_{n+1} < 3$.

Thus $2 < y_{n+1} < 3$, completing the induction.

Since $2 < y_n < 3$ for every n, the difference calculation applies at every step, so $y_{n+1} > y_n$ for all $n \geq 0$. The sequence is strictly increasing, and yet $y_n < 3$ for all n, as required.

The cobweb diagram shows what is going on: the curve $y = 5 - 6/x$ crosses the line $y = x$ at $x = 2$ and $x = 3$, lies above the line between them, and the iterates climb towards 3 without reaching it.



Checks

- At the fixed points the recurrences give constant sequences: $(3^2 + 6)/5 = 3$ and $5 - 6/3 = 3$, $5 - 6/2 = 2$, matching the factors $(t - 2)(t - 3)$.
- Starting from $y_0 = 2.5$: $y_1 = 2.6$, $y_2 \approx 2.692$, $y_3 \approx 2.771$, increasing and below 3; after 200 steps y_n is within 10^{-15} of 3. Starting from $x_0 = 3.1$: $x_1 = 3.122$, $x_2 \approx 3.149$, increasing.
- A computer tested both inductions with exact fractions for hundreds of random starting values, and confirmed that x_0 just below 3 gives $x_1 < x_0$, so the condition $x_0 > 3$ is needed.

Common slips

- Showing $x_{n+1} > x_n$ under the assumption $x_n > 3$ and stopping there, without proving by induction that every term exceeds 3.
- In part (ii), proving $y_{n+1} < 3$ but forgetting that $y_{n+1} > 2$ is also needed to continue the induction (it follows from $y_{n+1} > y_n$).
- Multiplying an inequality through by y_n without noting that $y_n > 0$.

Beyond STEP

Part (ii) is a bounded increasing sequence, and the *monotone convergence theorem* of analysis says such a sequence must converge; any limit L satisfies $L = 5 - 6/L$, so here $y_n \to 3$. In part (i), the sequence increases without bound, since a finite limit would be a fixed point greater than 3, and there is none. Whether iterates are attracted to or repelled from a fixed point depends on the slope of the map there, which is the start

of the theory of discrete dynamical systems. Abbott's *Understanding Analysis* treats monotone convergence carefully.

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